Worksheet 4-1 A Derivatives in Context #1
AP Calculus

Name SOLUTIONS

1. From the 2018 AP Exam:

t (years)	2	3	5	7	10
$H(t)$ (meters)	1.5	2	6	11	15

The height of a tree at time t is given by a twice-differentiable function H, where $H(t)$ is measured in meters and t is measured in years. Selected values of $H(t)$ are given in the table above.

(a) Use the data in the table to estimate $H'(6)$. Using correct units, interpret the meaning of $H'(6)$ in the context of the problem.

$H'(6) = \frac{H(7)-H(5)}{7-5} = \frac{11-6}{2} = \frac{5}{2}$ meters/yr

$H'(6)$ means the height of the tree is increasing at a rate of $5/2$ meters/year at $t=6$ years.

2. From the 2016 AP Exam (Calculator Active):

t (hours)	0	1	3	6	8
$R(t)$ (liters / hour)	1340	1190	950	740	700

Water is pumped into a tank at a rate modeled by $W(t) = 2000e^{-t^2/20}$ liters per hour for $0 \le t \le 8$, where t is measured in hours. Water is removed from the tank at a rate modeled by $R(t)$ liters per hour, where R is differentiable and decreasing on $0 \le t \le 8$. Selected values of $R(t)$ are shown in the table above. At time $t = 0$, there are 50,000 liters of water in the tank.

(a) Estimate $R'(2)$. Show the work that leads to your answer. Indicate units of measure.

$R'(2) = \frac{R(3)-R(1)}{3-1} = \frac{950-1190}{2} = -120 \frac{\text{liters/hr}}{\text{hour}}$

3. From the 2014 AP Exam (Calculator Active):

Grass clippings are placed in a bin, where they decompose. For $0 \le t \le 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.

(a) Find the average rate of change of $A(t)$ over the interval $0 \le t \le 30$. Indicate units of measure.

$\text{AROC} = \frac{A(30)-A(0)}{30-0} = -0.197$ pounds/day (or -0.196)

(b) Find the value of $A'(15)$. Using correct units, interpret the meaning of the value in the context of the problem.

$A'(15) = -0.164$ pounds/day

$A'(15)$ means the amount of grass clippings remaining in the bin is decreasing at a rate of -0.164 pounds/day at $t=15$ days

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4. From the 2017 AP Exam (Calculator Active):

When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t)\sin\left(\frac{t^3}{100}\right) \text{ for } 0 < t \le 12,$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

(b) Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.

$f'(7) = -8.119$ pounds per hour per hour

$f'(7)$ means the rate customers remove bananas from the display table is decreasing at a rate of 8.119 pounds per hour per hour at $t = 7$ hours

5. From the 2013 AP Exam (Calculator Active):

On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos\left(\frac{t^2}{18}\right)$, where t is measured in hours and $0 \le t \le 8$. At the beginning of the workday $(t = 0)$, the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \le t \le 8$, the plant processes gravel at a constant rate of 100 tons per hour.

(a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.

$G'(5) = -24.587$ tons per hour per hour

$G'(5)$ means the rate at which unprocessed gravel arrives at a gravel processing plant is decreasing at a rate of 24.587 tons per hour per hour at $t = 5$ hours

6. From the 2012 AP Exam (Calculator Active):

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F. The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

$$W'(12) = \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} = \frac{6.1}{6} = 1.016\,°F/\text{min}$$

$W'(12)$ means the temperature of water in a tub is increasing at a rate of 1.016°F/min at $t = 12$ minutes